

Validation Frameworks for Physics-Informed Machine Learning in Precipitation Hardening: Addressing Data Sparsity and Parameter Uncertainty

1. Introduction: The Epistemological Challenge of Sparse Data in Computational Metallurgy

The validation of computational models in materials science, particularly those simulating complex kinetic phenomena like precipitation hardening, has traditionally relied on the availability of dense, high-fidelity experimental datasets. However, the current research landscape is increasingly defined by the need to extract physical insights from sparse data—a constraint imposed by the high cost of electron microscopy and mechanical testing. The specific problem addressed in this report concerns the validation of a Kampmann-Wagner Numerical (KWN) precipitation model, implemented in Python using TensorFlow, which optimizes constitutive parameters against limited yield strength data (approximately 10 data points per experiment). This scenario represents a classic ill-posed inverse problem where the goal is to infer latent microstructural variables (precipitate size, number density, volume fraction) from a singular macroscopic property (yield strength) using a model with high degrees of freedom.

The core challenge lies not merely in fitting a curve to the data—modern optimization algorithms can trivially minimize the mean squared error (MSE) for 10 points using 5 parameters—but in proving that the optimized parameters possess physical fidelity. When training data is limited, the solution space often contains multiple local minima, meaning different combinations of interfacial energy, diffusion coefficients, and nucleation rates can mathematically reproduce the same yield strength curve despite implying contradictory microstructural states. This "equifinality" or "non-uniqueness" undermines the predictive capability of the model outside the training regime. Furthermore, the use of polynomial interpolation to augment this sparse data introduces mathematical artifacts (such as Runge's phenomenon) that can corrupt the physics-based gradients used during optimization.

This report establishes a comprehensive validation protocol tailored for these constraints, designed to withstand the scrutiny of a doctoral thesis defense. It argues that validation must move beyond deterministic error metrics and adopt a probabilistic framework. By integrating **Bayesian Inference** via TensorFlow Probability, one can quantify the uncertainty in the

constitutive parameters, transforming the limitation of "different coefficients for different experiments" into a statistical feature that characterizes material variability. The report also advocates for replacing polynomial interpolation with **Gaussian Process Regression (GPR)** to rigorously handle data sparsity. Finally, it provides detailed physical benchmarks for Al-Mg-Si alloys—the prototypical system for precipitation hardening—to serve as "ground truth" constraints for the model's internal variables.

The analysis proceeds by first deconstructing the KWN model and its limitations, then addressing the specific mathematical hazards of interpolation, followed by a detailed exposition of Bayesian calibration strategies, and concluding with a roadmap for multi-objective validation against literature benchmarks.

2. Theoretical Deconstruction of the KWN Model and Yield Strength Physics

To validate a computational model, one must first rigorously define the physics it approximates. The KWN model is a mean-field approach that discretizes the particle size distribution (PSD) to simulate the concurrent nucleation, growth, and coarsening of precipitates. Understanding the governing equations is essential to identifying where the "limited physics assumptions" mentioned in the query might lead to validation failures.

2.1 The Kampmann-Wagner Numerical Framework

The KWN model operates by solving the continuity equation for the number density $N(r, t)$ of particles with radius r at time t . Unlike classical analytical theories (like LSW theory) which apply only to the asymptotic coarsening regime, KWN covers the entire precipitation timeline.

The fundamental evolution equation governing the PSD is:

$$\frac{\partial N(r, t)}{\partial t} = -\frac{\partial}{\partial r}[v(r)N(r, t)] + J_{nuc}(t)\delta(r - r^*)$$

Here, $v(r)$ represents the growth rate of a precipitate, $J_{nuc}(t)$ is the nucleation rate, and the Dirac delta term accounts for the introduction of new stable nuclei at the critical radius r^* .

The growth rate $v(r)$ is typically derived from the steady-state diffusion approximation:

$$v(r) = \frac{dr}{dt} = \frac{C_m(t) - C_i(r)}{C_p - C_i(r)} \frac{D}{r}$$

Where $C_m(t)$ is the mean matrix concentration, C_p is the precipitate concentration (stoichiometric), $C_i(r)$ is the equilibrium concentration at the interface of a particle of radius r (governed by the Gibbs-Thomson effect), and D is the diffusion coefficient.

Validation Implication 1: Thermodynamic Consistency. The driving force for both nucleation and growth depends on the supersaturation of the matrix. A critical validation check is **mass conservation**. The code must track the solute budget:

$$C_{total} = C_{matrix}(t)(1 - f_v(t)) + C_{precipitate} f_v(t)$$

If the optimization minimizes yield strength error but violates this balance—for instance, by predicting a volume fraction f_v that requires more solute than is physically present in the alloy—the model is invalid.¹ The thesis must demonstrate that this constraint is either enforced (via Lagrange multipliers in the loss function) or monitored as a post-hoc verification metric.

2.2 The Physics of Yield Strength Superposition

The user’s code optimizes parameters against yield strength, a derived mechanical property. The relationship between microstructure and strength is non-linear and regime-dependent.

The total yield strength σ_y is generally modeled as a superposition of intrinsic lattice friction (σ_0), solid solution strengthening (σ_{ss}), and precipitation hardening (σ_p).

$$\sigma_y = \sigma_0 + \sigma_{ss}(C_m) + \sqrt{\sigma_p^2 + \sigma_{dis}^2}$$

The precipitation hardening term σ_p is the most complex, transitioning between a dislocation shearing mechanism (for small, coherent precipitates) and an Orowan bypassing mechanism (for large, incoherent precipitates).

Mechanism	Dependence on Radius (r)	Dependence on Vol. Fraction (f)	Physical Regime

Shearing	$\propto$	$\propto$	Under-aging (small coherent particles like GP zones, β'')
Bypassing (Orowan)	$\propto$	$\propto$	Over-aging (large incoherent particles like β' , β)
Transition	Peak Strength	Peak Strength	Critical radius r_c (typically 3-7 nm for Al-Mg-Si)

Validation Implication 2: The Degeneracy Problem.

This superposition creates a fundamental validation challenge known as **equifinality**. Different microstructural configurations can yield the same mechanical strength. For example, a microstructure with a high density of very small shearable particles might produce the exact same yield strength as a microstructure with a lower density of slightly larger particles near the peak-aging transition.

- If the user validates the code *only* against yield strength, they have not validated the precipitation physics; they have only validated the superposition model.
- The thesis must explicitly acknowledge this. A strong validation strategy involves "cross-property" validation: predicting a second property (like electrical conductivity, which depends linearly on solute concentration and is less sensitive to particle size) using the same parameters. If the model fits strength but fails conductivity, the internal variables (volume fraction vs. radius) are incorrect.³

2.3 The "Single Size Class" Assumption

Many efficient implementations of KWN, particularly those designed for rapid optimization in TensorFlow, simplify the full PSD into a "mean radius" model (or Single Size Class model). This assumes all precipitates at time t have the same radius $\bar{r}$ and number density N .

Validation Implication 3: Coarsening Failure. This assumption is physically perilous for the over-aging regime. Ostwald ripening (coarsening) is thermodynamically driven by the diffusion of solute from small particles (high solubility) to large particles (low solubility). In a single-class model, there is no size difference, so there is no thermodynamic driving force for coarsening. Such models often rely on empirical "coarsening rate" equations (e.g., LSW

scaling $r^3 \propto t$) rather than emergent physics.⁵

- **Thesis Defense:** The report must analyze whether the simulation code relies on a full PSD (multi-class) or a mean-radius approximation. If the latter, the validation section must justify this choice, likely by restricting the model's validity domain to the under-aging and peak-aging regimes where nucleation and growth dominate, and explicitly excluding deep over-aging where PSD width effects are critical.

3. Methodological Critique: The Dangers of Polynomial Interpolation and the Gaussian Process Alternative

The user notes that "a polynomial interpolation method has been used to augment the data limitation in training." This is a significant methodological weakness in the context of physics-informed machine learning. When data is sparse (10 points), the choice of interpolation scheme is not a neutral mathematical operation; it imposes a strong prior on the system behavior that can fundamentally corrupt the optimization.

3.1 The Artifacts of Polynomial Interpolation

Fitting a high-degree polynomial (e.g., degree 9 for 10 points) guarantees passing through every data point but invariably induces **Runge's phenomenon**—violent oscillations between data points, particularly at the boundaries. Even lower-degree polynomials (e.g., cubic splines) can introduce artificial inflection points that have no physical basis.

Physical Inconsistency:

Precipitation hardening curves are generally monotonic in specific regimes (monotonic rise to peak, monotonic fall). A polynomial interpolant might introduce a local minimum between two data points in the growth phase.

- **Gradient Corruption:** The KWN code, running in TensorFlow, likely uses gradient-based optimization (e.g., Adam or BFGS). These optimizers rely on the derivatives of the loss function. If the "target" interpolated curve has artificial wiggles, the optimizer will calculate gradients that try to force the physics model to reproduce these wiggles.
- **Result:** The optimizer will distort the physical parameters (e.g., creating a fluctuating nucleation rate) to chase numerical artifacts, resulting in a calibrated model that is physically nonsensical.⁷

3.2 The Gaussian Process Regression (GPR) Solution

To validate the code for a PhD thesis, it is strongly recommended to replace polynomial interpolation with **Gaussian Process Regression (GPR)**. GPR is a Bayesian non-parametric

approach that is standard in materials informatics for handling sparse datasets.⁹

Why GPR is Superior for this Application:

1. **Uncertainty Quantification:** Unlike a polynomial which gives a single (often wrong) value between points, GPR predicts a probability distribution. It provides a mean function $\mu(t)$ and a variance $\sigma^2(t)$. This variance naturally expands in the gaps between data points, representing the lack of knowledge.
2. **Physics-Compatible Priors (Kernels):** GPR uses a covariance function (kernel) to define the "smoothness" of the function. Using a **Matern 5/2** or **Radial Basis Function (RBF)** kernel enforces that the underlying function is continuous and differentiable, which aligns with the diffusion-controlled physics of precipitation. This avoids the unphysical oscillations of polynomials.¹⁰
3. **TensorFlow Integration:** Since the user's workflow is already in TensorFlow, implementing GPR is seamless using **TensorFlow Probability (TFP)**. The `tfp.distributions.GaussianProcessRegressionModel` class allows for exact inference on small datasets.

Feature	Polynomial Interpolation	Gaussian Process Regression
Output	Single deterministic value	Mean + Variance (Uncertainty)
Behavior in Gaps	Oscillations (Runge's Phenomenon)	Smooth reversion to mean/prior
Physical Priors	Implicit (degree of polynomial)	Explicit (Kernel length scale/smoothness)
Suitability for Optimization	Poor (corrupts gradients)	Excellent (weights loss by uncertainty)

3.3 Implementing Uncertainty-Weighted Optimization

With GPR, the validation strategy shifts from "matching the curve" to "maximizing the likelihood." The loss function in the TensorFlow optimization loop should be modified to weigh the error by the GPR uncertainty:

$$\mathcal{L}(\theta) = \sum_t \frac{(Y_{sim}(t, \theta) - \mu_{GPR}(t))^2}{2\sigma_{GPR}^2(t)} + \frac{1}{2} \log(2\pi\sigma_{GPR}^2(t))$$

Thesis Benefit: This approach tells the optimizer: "Fit the data tightly where we have measurements (low σ^2), but don't worry about matching the interpolation perfectly in the gaps (high σ^2).". This prevents the model from overfitting to interpolation artifacts, effectively solving Challenge #4.

4. Probabilistic Parameter Estimation: Bayesian Calibration for Sparse Data

The user identifies that "training data of each experiment will probably map to different trainable coefficients." In a deterministic framework, this is a failure mode—material constants like the diffusion pre-factor should not change arbitrarily between experiments. However, in a **Bayesian framework**, this variability is expected and can be rigorously modeled.

4.1 From Point Estimates to Posterior Distributions

Validating a model on sparse data requires abandoning the idea of finding "the" optimal parameter set. Instead, one must find the **posterior distribution** of parameters that are consistent with the data.

$$P(\theta|D) \propto P(D|\theta)P(\theta)$$

where θ is the vector of parameters (*param1 – 5*) and D is the experimental data.

Validation Strategy: Use **Markov Chain Monte Carlo (MCMC)** algorithms, such as Hamiltonian Monte Carlo (HMC) or the No-U-Turn Sampler (NUTS), provided by tfp.mcmc in TensorFlow Probability.¹²

- **Result:** instead of a single value for interfacial energy (γ), the analysis yields a probability density function.
- **Validation Criterion:** If the posterior distribution is extremely wide (effectively resembling the prior), it proves that the 10 data points contain **insufficient information** to constrain that specific parameter. This is a crucial validation result: it defines the limits of what the model *can* learn from the available data.¹⁴

4.2 Hierarchical (Multilevel) Bayesian Modeling

To address the variation of parameters across experiments (Challenge #2), the thesis should propose a **Hierarchical Bayesian Model** (HBM). HBM assumes that the parameters for each individual experiment θ_i are drawn from a global "population" distribution.¹⁵

- **Level 1 (Experiment):** $Y_i \sim \text{KWN}(\theta_i)$
- **Level 2 (Population):** $\theta_i \sim \mathcal{N}(\mu_{pop}, \Sigma_{pop})$
- **Priors:** $\mu_{pop}, \Sigma_{pop} \sim \text{Weakly Informative Hyper-priors}$

Why this validates the model:

1. **Shrinkage:** For experiments with very noisy or sparse data, the parameters θ_i will be "pulled" towards the global mean μ_{pop} . This regularizes the fit and prevents overfitting to the noise of a single bad experiment.
2. **Quantifying Heterogeneity:** The learned variance Σ_{pop} explicitly quantifies the material or process variability. If Σ_{pop} is small, the parameter is a true material constant. If it is large, there is uncontrolled experimental variation (e.g., quench rate differences). This turns a modeling "bug" (parameters varying) into a scientific "feature" (quantified process window).¹⁷

4.3 Sensitivity Analysis and Parameter Identifiability

With 5 parameters and limited data, some parameters will inevitably be "sloppy" (unidentifiable). For example, in the early stages of aging, the system might be sensitive to nucleation rate but insensitive to coarsening rate constants.

- **Validation Protocol:** Perform a **Sobol Sensitivity Analysis** or calculate the **Fisher Information Matrix**.¹⁸
- **Thesis Action:** Identify parameters with low sensitivity indices. Fix these parameters to literature values rather than letting the optimizer drift them to unphysical values. This reduces the dimensionality of the search space and improves the identifiability of the remaining "stiff" parameters.

5. Physical Validation: Benchmarking Against Literature Data

Since the user's proprietary data is limited, "transfer validation" against established literature

datasets is essential. The provided research snippets contain extensive data on **Al-Mg-Si (6xxx series)** alloys, which serves as the standard testbed for precipitation models. Validating the code against these public datasets lends credibility to the results obtained on the sparse proprietary data.

5.1 Quantitative Benchmarks for Al-Mg-Si Alloys

The thesis should benchmark the KWN code against specific, well-characterized datasets identified in the research snippets.

Benchmark 1: 6061-T6 Aging Curves Literature provides high-density yield strength vs. time data for AA6061 at various temperatures ($150^{\circ}C - 250^{\circ}C$).²⁰

- **Validation Test:** Can the model, calibrated on the $175^{\circ}C$ curve, accurately predict the $200^{\circ}C$ curve without re-training? This tests the validity of the activation energy terms (Q_a, Q_{sol}) encoded in the model. A simple curve-fitting algorithm (polynomial) would fail this extrapolation test; a physics-based KWN model should succeed.

Benchmark 2: Microstructural Kinetics (The "Internal State" Check)

Validating against strength is necessary but insufficient. The code must predict reasonable values for the internal state variables (radius, number density).

- **Data Source:** Snippets²⁴ provide quantitative TEM and SANS data for Al-Mg-Si.
 - **Precipitate Radius:** β'' needles typically grow from ~2 nm to ~50 nm length (radius ~1-3 nm) during peak aging.
 - **Number Density:** N_v typically starts at $10^{24}m^{-3}$ and drops to $10^{21}m^{-3}$ during coarsening.
- **Failure Mode:** If the model predicts a number density of $10^{15}m^{-3}$ (orders of magnitude off) to match the yield strength, the physics are broken. The thesis must explicitly check these magnitudes.

5.2 Physical Bounds for Constitutive Parameters

The optimization search space must be constrained by thermodynamics. Unbounded optimization often leads to "compensating errors" (e.g., unphysically high interfacial energy compensated by low diffusivity). The following table summarizes physical bounds for Al-Mg-Si derived from the research snippets, which should be used as **Bounded Priors** in the Bayesian calibration.

Parameter	Symbol	Physical Range (Al-Mg-Si)	Source
Interfacial Energy	γ	0.08 –	²⁷
Activation Energy (Diffusion)	Q_d	120 – (Mg/Si in Al)	²⁹
Solvus Enthalpy	ΔH_{so}	60 –	²⁹
Critical Radius (shearing/bypassing)	r_c	3 –	²⁸
Taylor Factor	M	3.0€ (isotropic fcc)	²⁸

Implementation Note: In TensorFlow, these can be implemented as tfb.Sigmoid bijectors that map the unbounded optimization space of the neural network to the bounded physical interval $[min, max]$.

6. Practical Validation Roadmap for the PhD Thesis

To structure the thesis validation chapter, the following roadmap transforms the challenges of data sparsity into a rigorous demonstration of advanced validation methodologies.

Phase 1: Algorithmic Verification (The "Unit Test")

Before touching experimental data, prove the code works on ideal data.

- Synthetic Data Generation:** Use the KWN code to generate a "perfect" aging curve using known parameters ($param_{tru}$).
- Synthetic Degradation:** Downsample this curve to 10 points and add Gaussian noise ($\sigma = 5$ MPa) to mimic the experimental constraints.
- Recovery Test:** Run the optimization on this synthetic sparse data.
 - Success Criterion:* Can the code recover $param_{tru}$ within the 95% confidence interval? If not, the inverse problem is mathematically unidentifiable with this data density, and no amount of experimental data will fix it.¹⁷

Phase 2: Data Robustification

Address the interpolation weakness.

1. **Implement GPR:** Replace polynomial interpolation with `tfp.distributions.GaussianProcessRegressionModel`.
2. **Kernel Selection:** Use a Matern 5/2 kernel to encode physical smoothness without enforcing infinite differentiability (which is too smooth for phase transitions).
3. **Weighted Loss:** modify the loss function to inversely weight residuals by the GPR variance.

Phase 3: Probabilistic Calibration (The "Main Event")

Calibrate against the experimental data.

1. **MCMC Sampling:** Run 4 chains of NUTS (No-U-Turn Sampler) for 2000 steps to map the posterior space of the 5 parameters.
2. **Convergence Check:** Use the Gelman-Rubin statistic ($\hat{R}$) to ensure the chains have converged.
3. **Result Reporting:** Report parameter distributions, not point estimates. Highlight correlations (e.g., strong correlation between γ and D) to demonstrate an understanding of the model's limitations.

Phase 4: Extrapolation and Cross-Validation

Prove predictive power.

1. **Leave-One-Curve-Out (LOCO):** If data exists for multiple temperatures, train on T_1 and T_3 , predict T_2 . This validates the Arrhenius terms.
2. **Physics Check:** Plot the predicted evolution of **precipitate radius** and **volume fraction**. Compare these curves qualitatively (and quantitatively if possible) against the literature bounds established in Section 5.2.

7. Conclusion

The validation of a complex physics-based simulation on sparse data is fundamentally an exercise in uncertainty quantification. By moving from deterministic fitting of polynomial-interpolated data to **probabilistic inference on raw data** (using GPR and Bayesian methods), the thesis can demonstrate a sophisticated handling of the "small data" problem.

The key to a successful defense lies in transparency: acknowledging the non-uniqueness of the inverse problem, rigorously bounding the parameters with literature values, and using

statistical methods to distinguish between what the data *actually* constrains (stiff parameters) and what it does not (sloppy parameters). This approach transforms the liability of limited data into a showcase of advanced ICME validation methodologies, proving that the candidate understands not just the code, but the epistemological limits of the simulation itself.

Table 1: Comparison of Interpolation Techniques for Sparse Materials Data

Method	Mathematical Basis	Artifact Risks	Suitability for Physics Models
Polynomial	$P(x) =$	High: Runge's phenomenon (oscillation at edges); creates artificial gradients.	Low: Oscillations induce unphysical driving forces in differential equations.
Spline (Cubic)	Piecewise polynomials	Medium: Can overshoot monotonic trends; sensitive to knot placement.	Medium: Better than high-order polynomials but assumes smoothness that may violate phase transition kinetics.
Gaussian Process	$f(x) \sim$	Low: Reverts to prior/mean in gaps; explicitly models uncertainty (σ^2).	High: Kernel choice (e.g., Matern) enforces physical continuity constraints; uncertainty weights the optimizer.

Table 2: Validation Hierarchy for the PhD Thesis

Level	Goal	Method	Metric for Success
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1. Verification	Ensure code solves equations correctly.	Mass balance check; Synthetic data recovery.	Solute conservation < 0.1% error; Recovery of $param_{tru}$ from synthetic data.
2. Calibration	Fit parameters to sparse data with UQ.	Bayesian Inference (MCMC/HMC) via TensorFlow Probability.	Posterior distributions converge ($\hat{R} \approx 1$); Posterior $\neq$ Prior (data is informative).
3. Validation	Prove physical predictive power.	Leave-One-Curve-Out (LOCO) CV; Comparison to Literature TEM data.	Prediction of held-out temp curve within 95% CI; r and N_v within order of magnitude of literature.

Table 3: MCMC Algorithm Selection for KWN Model Calibration

Algorithm	Mechanism	Pros	Cons	Recommendation
Metropolis-Hastings	Random Walk	Simple to implement.	Slow convergence for high-dim, correlated parameters.	Avoid for 5-param physics models.
Hamiltonian MC (HMC)	Gradient-based physics analogy	Efficient exploration of complex posteriors.	Requires tuning of step size/leapfrog steps.	Good , natively supported in TFP.

NUTS (No-U-Turn)	Adaptive HMC	Auto-tunes HMC parameters; highly efficient.	Computationally expensive per step.	Best Choice for rigorous thesis results.
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